RAJALAKSHMI ENGINEERING COLLEGE

ABINAVVA 251001504-2025-IT

A2

Started on	Tuesday, 22 September 2026, 9:35 AM
State	Finished
Completed on	Tuesday, 22 September 2026, 9:38 AM
Time taken	2 mins 7 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main(){
    int n,x,l=0,r,m,ans=-1;
    scanf("%d",&n);
    int a[n];

    for(int i=0;i<n;i++) scanf("%d",&a[i]);
    scanf("%d",&x);

    r=n-1;
    while(l<=r){
        m=(l+r)/2;
        if(a[m]<=x) ans=a[m],l=m+1;
        else r=m-1;
    }

    printf("%d",ans);
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔

	Input	Expected	Got	
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course